Amit Chandran

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I am a Mechanical Engineer with skills and experience in development of software, hardware and mechanical systems within cross-functional teams. Aspiring to work within robotics, control or embedded systems.

Education

UCL - MEng Mechanical Engineering **Expected June 2028**

- Relevant Modules: Thermodynamics, Dynamics, Mathematical Modelling, Robotics, Control Systems
- Current Average: 78% - First Class

Experience

Systems Engineering Intern – Leonardo UK **Summer 2025 and 2026**

- Operated as a systems engineer, coordinating 10 other interns to seamlessly integrate antenna, firmware, hardware and software components into a continuous RF to web-server pipeline.
- Implemented HDL-compatible signal processing pipelines in Simulink and deployed them on an RFSoC FPGA board, producing high-resolution spectra
- Developed a compression algorithm reducing FFT data bandwidth requirements by over **2500×**, minimising network utilisation whilst preserving critical information
- Optimised signal max-hold algorithm using MATLAB by reducing complexity from O(N^2) to O(N) achieving a **99.3%** simulation time reduction

Projects / Extracurricular

Technical Lead and Drone Controls Engineer – UCL Rover Team **January 2025 – present**

- Programmed an edge-computing vision pipeline in Python and OpenCV on a Raspberry Pi to track ArUco markers in real-time, with an average pose error of $<$**1cm**
- Established a robust UDP telemetry link to stream positional data seamlessly from the edge device to a high-level MATLAB/Simulink control model
- Built a UART-based MAVLink telemetry architecture connecting Simulink models directly to an H743 flight controller for real-time flight control

Smart Pet Feeder – BLE IoT System **October - March 2026**

- Developed C++ firmware on an ESP32-S3, interfacing a NEMA17 stepper motor and HX711 strain-gauge amplifier for real-time mass measurement and portion control
- Implemented an event-driven BLE GATT server (Nordic UART Service) that dynamically synced ESP32 system time via an Android client, eliminating external RTC chip costs
- Engineered a tuneable control loop allowing the device to dispense user-defined portion sizes at pre-set times, achieving **±1%** mass accuracy across 20–100g portions and sub-2s timing precision in a real world demo
- Deployed firmware and build instructions to an open source repository, making use of DfMA (Design for manufacturing and assembly) to allow the public to easily modify and build to a use case

IMechE Design Challenge – Autonomous Robotic Device **January – March 2025**

- Led the team to develop a bespoke electro-mechanical solution without programmable microcontrollers, maximising autonomy over competitors and placing 3rd of 45 teams
- Designed discrete MOSFET logic gate circuitry with a H-bridge on a custom EasyEDA PCB, cutting down on moving components whilst improving repeatability
- Formulated a rigorous test strategy and built a test rig to rapidly implement upgrades to the vehicle, reducing mass by **85%** and wire length by **1.2m**

Skills

- **Programming Languages** – MATLAB, Python, C++
- **Tools / Software** – Simulink, Fusion 360, Git, Azure DevOps, EasyEDA
- **Manufacturing** – 3D Printing, Laser Cutting, Standard workshop equipment, Soldering